LEAN: A Lightweight and Scalable Attention Network for ECG Age Estimation as a Marker of Cardiovascular Health

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Abstract

An electrocardiogram (ECG) is a widely used non-invasive test for diagnosing cardiac disease. Recent work has established that the gap between a neural-network-derived ECG age and the chronological age is a useful measure of cardiovascular well-being. However, most ECG age estimators reuse networks built for disease diagnosis, and these tend to be computationally heavy and difficult to scale. We propose LEAN, a lightweight and scalable hybrid of convolution and self-attention for efficient ECG age estimation. LEAN extracts features through a shallow backbone of ECGNext blocks and then applies a self-attention block that supplies the long-range receptive field the shallow backbone cannot reach on its own. A multi-shot inference scheme further refines each prediction by averaging over several crops of the recording. Furthermore, a healthy-reference analysis shows that LEAN-derived ECG age gaps provide a disease-associated aging marker, with pathological recordings appearing approximately four years older than healthy recordings across multiple cardiac conditions. LEAN is instantiated as a family of variants, so a practitioner can match the model to the available computational budget. Its smallest variant, LEAN-Lite, surpasses the strongest existing ECG age estimation network by 4.3% in MSE while using 42× fewer parameters, and the family as a whole attains accurate age estimation at a small fraction of the memory and computation of prior models, making it deployable on resource-constrained point-of-care devices for cardiovascular screening.

1. Introduction

Deep learning now achieves state-of-the-art performance on individual cardiac diagnoses, but each model remains bound to the labels it was trained on, limiting its ability to generalize across cardiovascular conditions. This limitation matters because cardiovascular disease develops gradually, and conditions often overlap or progress into one another, as when hypertension promotes atherosclerosis and eventually coronary artery disease. Clinicians, therefore, need an interpretable marker of overall cardiovascular health rather than a verdict on one predefined condition at a time [1, 2].

ECGs are well-suited to provide such a marker, because their waveform changes with aging, latent cardiovascular disease, and genetic variation [3, 4, 5]. Neural networks

can learn these age-related and disease-related waveform patterns directly from raw ECGs [6, 7]. ECG age converts those waveform changes into a single interpretable measure of cardiovascular status. It estimates the age implied by the ECG and compares it with chronological age, and the resulting gap, or delta age, carries the clinically useful signal [8]. Prior studies have linked a larger ECG age gap to mortality and broader disease burden, which makes ECG age a practical indicator of overall cardiovascular health [9, 10, 11, 12, 13].

Work on ECG age estimation has converged on end-to-end deep learning, with validation extending across increasingly diverse datasets and populations. Zvuloni *et al.* [14] compared feature engineering, end-to-end deep learning, and a hybrid of the two, and showed that hand-crafted features alone are far less effective than an end-to-end model. Attia *et al.* [15] had earlier introduced AttiaNet, a 1-D CNN that estimates age from raw 12-lead recordings in the Mayo Clinic database of a North American population. Strodthoff *et al.* [16] benchmarked several architectures on the public PTB-XL dataset [17] and reported lower estimation error for normal subjects than for abnormal ones, consistent with Attia *et al.* [15]. Lima *et al.* [13] adapted a deep 1-D residual network (ResNet1D) to a Brazilian population and validated it on external South American databases, where subjects whose ECG age exceeded their chronological age by more than eight years showed higher mortality.

Despite this progress, the ECG age metric has yet to gain traction in routine clinical practice or day-to-day use, mainly because no purpose-built ECG age network is scalable and lightweight. A scalable architecture can be reconfigured across a range of sizes, which lets practitioners trade computation for accuracy as their hardware allows. A lightweight architecture, in turn, keeps the parameter count and storage footprint small, so that it is deployed on the constrained hardware of ECG machines and wearables and runs cheaply on a population scale for longitudinal validation. The main goal of this work is to deliver a scalable and lightweight ECG age network.

To close this deployment gap, we introduce the Lightweight ECG Attention Network (LEAN) for ECG age estimation. LEAN pairs a shallow, scalable ECGNext-based backbone with self-attention, multi-shot inference, and healthy-reference age-gap analysis. Together, these components keep the model lightweight while preserving the ability to capture local morphology, longer-range dependencies, and pathology-associated shifts in ECG-derived age. Beyond efficiency, we also ask whether the resulting age estimate carries clinical meaning on its own, and we show that it can flag disease-associated aging without being trained on disease labels.

Our main contributions are as follows:

- We design a shallow and scalable convolutional backbone from the *ECGNext block*, which extracts ECG features effectively while reducing the parameter burden of deeper layers.
- We add self-attention over convolutional features to capture long-range dependencies without relying on a deeper backbone.
- We propose a *multi-shot inference* scheme, enabled by random-crop training, that averages predictions across several fixed-length ECG crops to improve age estimation.
- We introduce a healthy-reference age-gap analysis that corrects for age-dependent regression bias and quantifies pathology-associated aging without disease-specific training.

Stage	Output ($C \times L$)	Layer specification
Input	$12 \times T$	12-lead ECG segment, T samples
Network Stem Temporal Block	$C_1 \times \frac{T}{4}$	[Conv1D, k BatchNorm1D ReLU MaxPool, P] $\times 2$
Deep Feature Extraction ECGNext Block	$C_2 \times \frac{T}{16}$	[Depthwise Conv1D, C LayerNorm 1×1 Conv, $2C$ (expand) GELU LayerNorm 1×1 Conv, K (project) MaxPool, P] $\times 2$
Global Feature Reweighting Self-Attention	$C_2 \times \frac{T}{16}$	Multi-head self-attention, h heads
Head	1	Global average pooling $\rightarrow$ Linear $\rightarrow$ age

Figure 1. Overview of the proposed LEAN architecture with stages, output dimensions, and the sublayers within each block.

2. Methodology

This section presents the proposed LEAN architecture in detail. It first gives an overview of the network and then describes its two main components, namely the lightweight ECGNext backbone that extracts age-relevant ECG features and the self-attention block that reweights those features globally.

2.1. LEAN Framework

2.1.1 Overview

The goal of LEAN is to deliver accurate ECG age estimation while keeping the parameter count and storage footprint small and remaining scalable across computational budgets. The stages column in Figure 1 illustrates the roles of its components, namely the network stem, and global feature reweighting. The 12-lead ECG first passes through two temporal blocks that perform preliminary feature extraction. A sequence of ECGNext blocks then produces deeper, age-relevant features. The resulting convolutional feature maps are reweighted globally by the self-attention

block, which lets every position attend to every other and thereby captures the long-range dependencies that the shallow backbone cannot. The reweighted feature maps are finally condensed by global average pooling and passed to a linear layer that estimates the ECG age.

2.1.2 LEAN Backbone

The LEAN backbone is deliberately shallow. It passes the 12-lead ECG through a stem of two temporal blocks and then through two ECGNext blocks, our one-dimensional adaptation of the ConvNeXt block [18], which perform the deeper feature extraction. Layer specifications in Figure 1 shows the design of both block types. The stem is built from standard 1-D convolutions. We use standard rather than depthwise separable convolution here because it captures low-level features such as peaks and troughs by processing all channels jointly, which lets the stem learn spatial and cross-channel correlations at the same time. Depthwise separable convolutions instead suit the ECGNext stage, where the spatial resolution has fallen and the channel count has grown, since they lower the parameter count with little loss in representational power. The ECGNext block follows the vision transformer and separates spatial and channel mixing into two stages. A depthwise convolution with a large kernel first mixes information along the signal and widens the receptive field. The feature maps are then layer-normalized, again as in the vision transformer. Pointwise convolutions finally mix the channels through an inverted-bottleneck design. MobileNetV2 [19] and EfficientNet [20] quadruple the channels inside this bottleneck, whereas the ECGNext block only doubles them, which keeps the parameter count low. The block uses the Gaussian Error Linear Unit (GeLU) in place of ReLU for a smoother nonlinearity and closes with a max-pooling step that reduces the spatial dimension of the feature map. Ignoring bias terms, the parameter counts of the convolutional layers in the temporal block and the ECGNext block are

$$\text{ECGNext Block} = f\, n_{\text{in}} + 2n_{\text{in}}^2 + 2n_{\text{in}}n_{\text{out}} \qquad (1)$$

$$\text{Temporal Block} = f\, n_{\text{in}}\, n_{\text{out}} \qquad (2)$$

where f is the 1-D convolutional kernel size, n_{in} is the number of input channels, and n_{out} is the number of output channels. Comparing Equations 1 and 2 shows that the ECGNext block replaces a product of three factors with a sum of terms in only two factors, because it mixes space and channels separately. The ECGNext block therefore uses progressively fewer parameters than the temporal block as the channel counts (n_{in} and n_{out}) and the kernel size (f) grow, which is the source of LEAN's scalability. Keeping the backbone shallow compounds these per-block savings. It also avoids a known failure mode, since deep networks for ECG age estimation are prone to overfitting and can perform worse than smaller ones [21], which is why we limit the backbone to two blocks of each type.

Table 1. Channel configuration of the four LEAN variants. Each entry is the number of channels at that stage, and the feature dimension for the self-attention block. The kernel size k and output length appear in the header; every stage uses a pooling factor of two, the self-attention block uses four heads in all variants, and each variant ends in an output perceptron for age estimation.

Variant	Temporal 1 k=7, $L/2$	Temporal 2 k=5, $L/4$	ECGNext 1 k=5, $L/8$	ECGNext 2 k=3, $L/16$	Self-Attn. (dim), $L/16$
Lite	16	16	32	32	32
Base	32	32	64	64	64
Plus	48	48	96	96	96
Max	64	64	128	128	128

2.1.3 Self-Attention Block

The convolutional layers in the ECGNext backbone process the ECG with fixed-size kernels, so the receptive field of the network is static and bounded by the kernel size and the number of layers. Because the backbone is deliberately shallow, it cannot, on its own, relate distant parts of the signal. The self-attention block removes this limitation by reweighting the convolutional features globally, allowing every position in the feature map to draw on every other position.

Let $X \in \mathbb{R}^{L\times D}$ be the convolutional feature map produced by the backbone, where L is the sequence length and D the number of channels. The block first projects X into queries, keys, and values through learned linear maps,

$$Q = XW^Q, \quad K = XW^K, \quad V = XW^V, \qquad (3)$$

with $W^Q, W^K, W^V \in \mathbb{R}^{D\times D}$. It then measures how strongly each position relates to every other by comparing their queries and keys, which produces a matrix of pairwise scores,

$$S = \frac{QK^\top}{\sqrt{D}}, \qquad A = \text{softmax}(S), \qquad Z = AV. \qquad (4)$$

The scores are normalized row-wise by a softmax to produce attention weights, which are used to compute each output feature as a weighted sum of value vectors. Thus, Equations 4 transform the local morphology features extracted by the backbone into a global reweighted representation by modeling long-range dependencies in the ECG signal. In practice, projections are divided into h parallel heads and concatenated, allowing the block to capture multiple relationship patterns simultaneously.

2.1.4 LEAN Framework Variants

LEAN is scalable by design, and we instantiate it as four variants, Lite, Base, Plus, and Max, that differ in parameter count. The variants are obtained by adjusting the number of

channels in the ECGNext backbone and the self-attention block. Table 1 gives their detailed configurations.

3. Experimental Setup

This section details how LEAN and the competing networks are trained and evaluated. It first describes the ECG datasets, then the implementation and training protocol, including the random-crop training and multi-shot inference schemes, and finally the metrics that quantify estimation accuracy and computational cost.

3.1. Dataset Description

PTB-XL [17] contains 21,837 clinical 12-lead recordings of 10 seconds each, acquired from 18,885 patients, and it serves as our primary corpus and a common benchmark for end-to-end ECG age estimation [16]. Two cardiologists annotated every recording under the SCP standard for form, rhythm, and diagnostic statements, producing five coarse classes that further divide into twenty-four subclasses. The dataset also supplies per-record metadata on age, sex, height, weight, acquisition site, signal noise, and cross validation folds.

The Chapman dataset [22] adds 12-lead recordings from 10,646 patients sampled at 500 Hz, which we downsample to 100 Hz for age estimation. Clinical experts labeled each recording for eleven common rhythms and sixty-seven diagnostic conditions. It was compiled jointly by Chapman University and Shaoxing People's Hospital to support large-scale computational study of cardiovascular signals.

3.2. Implementation Details

All networks were implemented in PyTorch 1.9 and trained on 100 Hz signals, a low sampling rate chosen to reduce the computational budget and to emulate resource-constrained deployment. Optimization minimized the mean squared error over 200 epochs with a batch size of 96 and an initial learning rate of 0.001, and a scheduler cut the learning rate by a factor of ten every 75 epochs. For each run, the weights with the lowest validation error were kept for testing. Every network was assessed under ten-fold cross-validation with the folds defined in the PTB-XL metadata, and all reported numbers are averages over the ten folds, which prevents any single portion or class of the data from skewing the outcome. To examine behavior on under-represented ages, we additionally evaluated the networks on a **few-shot** subset, comprising the ages that appear fewer than forty times in the dataset.

3.2.1 Random Crop and Multi-Shot Inference

During training, each recording is reduced to a random 8-second crop, so LEAN sees a different view of the signal at every epoch. This improves generalization and makes the model tolerant of the systematic and random corruptions that occur during acquisition. Eight seconds is sufficient because crops of this length have been shown to match the age-estimation accuracy of the full 10-second recording [21]. This design, however, restricts inference to a single fixed 8-second segment, leaving a substantial portion of the temporal information unexploited. To address this limitation, we adopt a multi-shot inference strategy that extracts ten overlapping 8-second segments from the complete ECG recording. Each segment is processed independently to obtain an age estimate, and the final prediction is computed as the arithmetic mean of all segment-level estimates (Figure 2): $\tilde{y} = \frac{1}{M}\sum_{j=1}^{M} f(s_j)$, where s_j denotes the j-th 8-second segment extracted from the complete ECG recording S, $f(s_j)$ represents the predicted age for that segment, and $M = 10$ is the total number of inference segments.

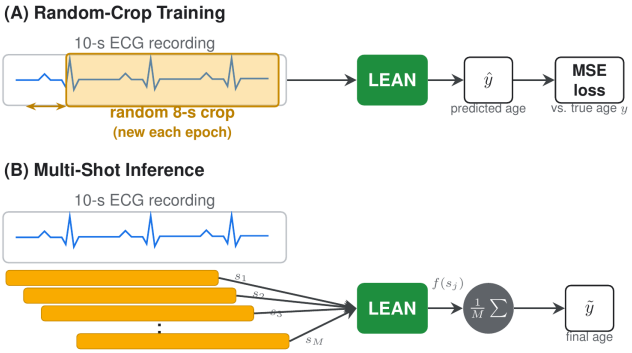


Figure 2. Random crop training and multi-shot inference.

3.3. Performance Metrics

Each network is judged both on how closely its estimates track the true age and on the computation and memory it demands. Accuracy rests on three error measures. The mean absolute error (MAE) takes the mean of the absolute gaps between the predicted and true age, which makes it directly interpretable in years and resistant to outliers, $\text{MAE} = \frac{1}{n}\sum_{i=1}^{n}|y_i - \hat{y}_i|$. The error geometric mean (GM) multiplies these absolute gaps and takes the n-th root, reporting their typical size on a relative scale while suppressing extreme values even more than the MAE, $\text{GM} = \left(\prod_{i=1}^{n}|y_i - \hat{y}_i|\right)^{1/n}$. The mean squared error (MSE) squares each gap before averaging, and its smoothness is the reason the networks are trained to minimize it, $\text{MSE} = \frac{1}{n}\sum_{i=1}^{n}(y_i - \hat{y}_i)^2$. In every case, n is the number of test samples, y_i the chronological age, and $\hat{y}_i$ the estimated age.

Resource use is assessed through parameter count, disk footprint, multiply-accumulate operations, VRAM usage, and inference times, capturing model size, computational cost, GPU memory demand, and prediction speed.

Table 2. Age-estimation accuracy on PTB-XL, organized by model family. Each entry gives the mean with the standard deviation in parentheses across 10 cross-validation folds, and lower is better for every metric. Scores are reported separately on the full test set (All) and the few-shot subset (Few). Bold marks the best value in each column.

		Full test set (All)			Few-shot subset (Few)		
Family	**Network**	**MSE**	**MAE**	**GM**	**MSE**	**MAE**	**GM**
Proposed	LEAN-Lite	106.35 (4.47)	7.98 (0.18)	5.18 (0.15)	322.27 (65.57)	15.85 (1.36)	13.74 (1.23)
	LEAN-Base	102.42 (4.13)	7.80 (0.18)	5.03 (0.18)	315.09 (63.77)	15.35 (1.31)	12.68 (1.78)
	LEAN-Plus	101.92 (3.44)	7.80 (0.14)	5.07 (0.12)	318.12 (67.58)	15.50 (1.36)	13.28 (1.24)
	LEAN-Max	**101.79 (6.95)**	**7.75 (0.25)**	**5.02 (0.21)**	320.09 (70.50)	15.36 (1.28)	12.70 (1.64)
Prior ECG	AttiaNet [15]	111.13 (3.77)	8.17 (0.13)	5.26 (0.16)	360.54 (60.64)	16.92 (1.36)	14.89 (1.49)
	ECG12Net [23]	117.22 (7.30)	8.43 (0.23)	5.50 (0.22)	307.83 (47.58)	15.32 (1.31)	12.42 (1.64)
	ResNet1D [13]	150.17 (20.08)	9.42 (0.65)	6.15 (0.55)	**254.79 (82.54)**	**11.68 (1.19)**	**7.50 (0.79)**
Lightweight	MobileNetV2 [19]	112.76 (5.22)	8.21 (0.22)	5.29 (0.21)	322.24 (80.14)	15.63 (2.00)	12.99 (2.69)
	EfficientNetB0 [20]	109.08 (4.43)	8.11 (0.13)	5.31 (0.13)	335.41 (98.62)	16.08 (2.00)	13.98 (1.81)

Table 3. Computational footprint of every evaluated architecture on PTB-XL, laid out by resource metric (rows) across the three model families (columns). All figures use single-sample CPU inference. Bold marks the most efficient network for each metric.

	Proposed LEAN variants				Prior ECG networks			Lightweight vision	
Resource metric	**Lite**	**Base**	**Plus**	**Max**	**AttiaNet** [15]	**ECG12Net** [23]	**ResNet1D** [13]	**MobileNetV2** [19]	**EfficientNetB0** [20]
Inference time, CPU (s)	1.249	2.27	3.13	4.305	**0.878**	82.71	61.57	18.21	24.83
Parameter count	**14,121**	51,281	111,481	194,721	593,505	5,668,666	6,941,985	2,225,153	4,008,829
Disk utilization (MB)	**0.06**	0.21	0.45	0.78	2.37	22.67	27.77	8.9	16.04
Est. VRAM w/ input (MB)	**1.11**	2.26	3.51	4.84	3.08	89.35	41.61	40.75	49.25
Multiply–add ops (M)	**2.13**	5.6	10.39	16.52	3.7	540.65	1640	116.48	160.07

4. Results & Discussion

This section evaluates LEAN from the perspectives of predictive performance and computational efficiency. We first compare LEAN variants with existing ECG age-estimation methods on PTB-XL. We then examine the performance vs efficiency trade-off relative to lightweight ECG architectures to assess scalability. Finally, we train and evaluate on ChapmanECG dataset to validate the generalizability of LEAN on different demographic.

4.1. Performance and Efficiency of LEAN

Table 2 and Table 3 place LEAN alongside the networks from the literature in both accuracy and computational cost. On the full test set, LEAN is the strongest network on every metric. LEAN-Max records the lowest MSE (101.79), MAE (7.75), and GM (5.02), and even LEAN-Lite, the smallest variant, already surpasses the best dedicated network, AttiaNet, together with the general-purpose EfficientNetB0 and MobileNetV2. The ranking holds across three metrics of different character (MSE, MAE, and GM), which indicates that the improvement reflects a genuinely better fit rather than an artifact of any single measure. The benefit of scaling is front-loaded. The step from LEAN-Lite to LEAN-Base lowers the MSE from 106.35 to 102.42, whereas the two further steps to LEAN-Max recover only about half a point in total. LEAN-Base therefore reaches within 0.6 MSE of the best configuration at roughly a quarter of its parameters, and the mid-range variants are also the most stable across folds, with standard deviations of 4.13 and 3.44 for Base and Plus against 6.95 for Max. These variants make attractive defaults when stability or storage is the priority.

The accuracy comes at a footprint smaller by orders of magnitude. LEAN-Lite holds the lowest value in every resource column of Table 3, at 14,121 parameters, a 0.06 MB disk footprint, 1.11 MB of VRAM, and 2.13 million multiply-accumulate operations. The gap to the large diagnostic networks is stark. LEAN-Lite performs roughly 250 times fewer operations than ECG12Net and about 770 times fewer than ResNet1D, and it uses around 80 times less VRAM than ECG12Net. Even the largest variant stays compact, since LEAN-Max occupies under one megabyte on disk, so the entire family fits on storage-limited hardware. On CPU, AttiaNet is marginally faster than LEAN, at 0.878 s against 1.2 to 4.3 s, because it uses neither attention nor test-time averaging, yet both run in the low-second range while ECG12Net and ResNet1D need 83 and 62 seconds and are effectively unusable at the edge. Scaling LEAN does raise cost, as the wider feature maps of the larger variants enter the self-attention block at a higher dimension and push their VRAM and multiply-accumulate counts above AttiaNet, but these counts remain far below every large model in the table.

The subset results are more nuanced. LEAN leads comfortably on the full test set, yet the few-shot group, which holds the rare and under-represented ages, is won by the two largest models. ResNet1D takes it on all three metrics and

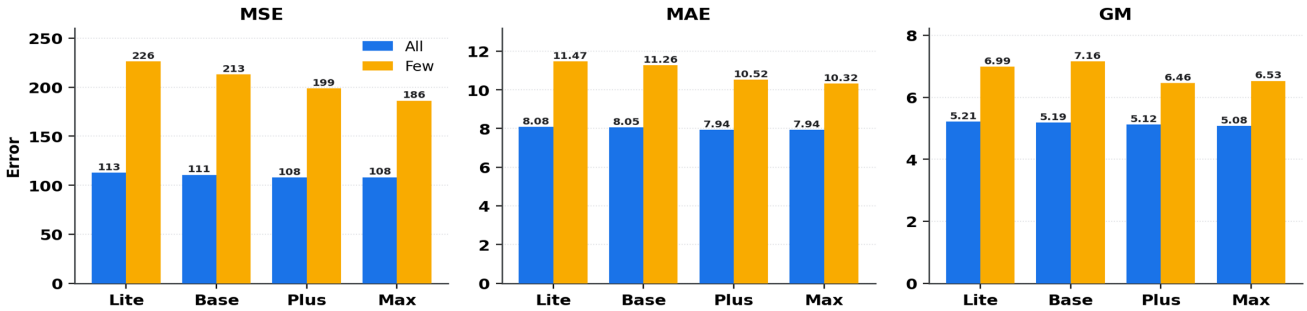


Figure 3. MSE of LEAN architecture on the ChapmanECG dataset test set (All) and instances with few counts (Few) in training set.

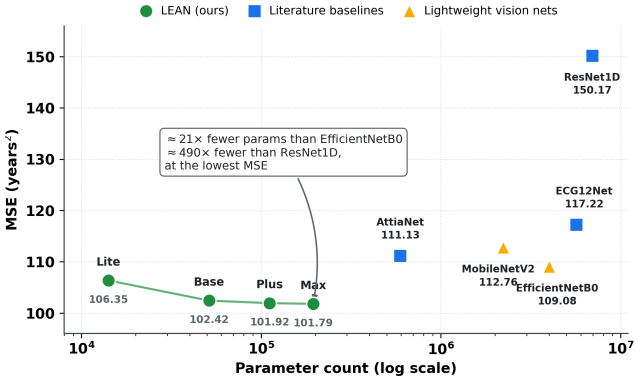


Figure 4. Performance vs Parameters tradeoff of the LEAN relative to the networks in the literature and lightweight architecture.

ECG12Net follows, while LEAN and the remaining networks trail. This ordering ties few-shot accuracy closely to parameter count, because a network with millions of weights can devote spare capacity to fitting the sparse tail of the distribution. LEAN is not the weakest in this group and still improves on AttiaNet and on the general-purpose models, but with only tens to hundreds of thousands of parameters it cannot match the raw capacity of ResNet1D. LEAN instead concentrates its representation on the densely populated center of the age distribution, where most subjects lie, and accepts a modest loss at the extremes. Figure 4 summarizes this balance by plotting the accuracy of LEAN against its size relative to the other networks in the domain.

4.1.1 Comparison with Lightweight Networks

Lightweight models built for other domains are frequently reused as ready-made efficient backbones, so we also compare against MobileNetV2 [19] and EfficientNetB0 [20], both designed for 2-D image tasks. The 2D functions within these networks (e.g., Conv, Batchnorm, etc.) were replaced with their 1D counterparts. On ECG age estimation, MobileNetV2 reaches an MSE of 112.76 and an MAE of 8.21, and EfficientNetB0 is slightly better at 109.08 and 8.11. These numbers are on par with AttiaNet, which matches them at a far lower resource cost because it is tailored to one-dimensional signals. LEAN improves on the general-purpose models and on AttiaNet in accuracy while staying more efficient than either, which underlines the benefit of an architecture designed specifically for ECG age estimation rather than borrowed from image recognition.

4.2. Generalization to the ChapmanECG Dataset

To test whether the architecture generalizes beyond a single population, we train and evaluate LEAN on the ChapmanECG dataset [22], whose subjects are predominantly of Chinese ethnicity and thus demographically distinct from PTB-XL. Figure 3 reports the performance on the entire test set and fewshot age ranges. LEAN-Lite reaches an MSE of 112.89 and LEAN-Max lowers it to 108.18, so the scaling behavior observed on PTB-XL reappears on the new population. These errors sit close to the PTB-XL results, which indicates that the architecture adapts to the dataset rather than relying on one cohort. The modest residual gap is most plausibly a sampling-rate effect: the Chapman recordings are supplied at 400 Hz and downsampled to 100 Hz, a step that can blunt sharp features such as the R-peak and slightly lower accuracy. Together, these results show that LEAN is a suitable and adaptable framework for ECG age estimation across diverse datasets.

4.3. The healthy-reference age gap as a marker of cardiac pathology

We ask whether the learned age representation is sensitive to cardiac pathology. To this end we train (folds 1-8) LEAN on healthy recordings only (i.e., diagnostic superclass is NORM). We call the result the *healthy-reference* model and apply it to the full test fold (fold 10). For a recording of chronological age a and prediction $\hat{a}$, the *age gap* is $g = \hat{a} - a$. Pathological morphology reproduces several electrical signatures of aging. A model that has seen only healthy hearts should therefore over-estimate the age of a diseased recording. Its gap should be positive. Its magnitude should track the degree of disease-induced aging.

The composition of the test fold complicates this analysis (Table 4). The dataset is strongly skewed toward older subjects, with only 83 recordings in the twenties compared with 481 and 452 in the sixties and seventies. Healthy and diseased subjects also occupy different regions of the age distribution. Healthy recordings peak in the fifties, and only 19 of 878 subjects are older than 80. In contrast, MI contains

Table 4. Age composition of the test fold by diagnostic superclass (shading $\propto$ count).

Class	20s	30s	40s	50s	60s	70s	80s	Total
NORM	70	114	157	217	185	116	19	**878**
STTC	4	15	23	75	123	137	125	**502**
MI	0	6	31	82	113	166	139	**537**
HYP	2	6	15	28	77	72	55	**255**
CD	9	18	33	71	95	125	121	**472**
All	**83**	**153**	**237**	**404**	**481**	**452**	**309**	**2119**

Table 5. Age-matched ECG-age gaps for the four disease superclasses. Per-decade raw gaps and healthy-adjusted deltas (Δ) are reported, together with the count-weighted pooled Δ over ages 40–79. †: 95% bootstrap CI includes zero.

Class		40s	50s	60s	70s	Pooled Δ (95% CI)
STTC	raw	+9.9	+6.1	+1.8	−5.1	
	Δ	$5.3_{(23)}$	$4.1_{(75)}$	$4.3_{(123)}$	$3.7_{(137)}$	+4.1 [3.0, 5.2]
MI	raw	+14.5	+6.5	+1.5	−4.8	
	Δ	$9.9_{(31)}$	$4.5_{(82)}$	$4.1_{(113)}$	$4.0_{(166)}$	+4.6 [3.6, 5.7]
HYP	raw	+8.5	+5.6	+0.6	−5.6	
	Δ	$3.9_{(15)}{}^{\dagger}$	$3.7_{(28)}{}^{\dagger}$	$3.2_{(77)}$	$3.2_{(72)}$	+3.3 [1.9, 4.7]
CD	raw	+10.4	+7.4	+1.6	−4.7	
	Δ	$5.8_{(33)}$	$5.4_{(71)}$	$4.1_{(95)}$	$4.1_{(125)}$	+4.6 [3.4, 5.7]
NORM	raw	+4.6	+2.0	−2.5	−8.8	age bias removed by Δ

no recording below 30, while 305 of its 537 recordings are older than 70. This imbalance introduces an age-dependent regression bias. Predictions are pulled toward the training-age centroid, yielding positive age gaps for younger subjects and negative gaps for older subjects. A pooled analysis would therefore confound disease effects with age bias. We instead perform decade-stratified comparisons, evaluating each disease class against age-matched healthy subjects.

We report three quantities for each disease class c and age decade d. The *raw gap* $\bar{g}_{c,d}$ is the mean signed prediction error for recordings of class c within decade d, and therefore retains the age-dependent regression bias. To remove this effect, we compute the *age-matched delta*: $\Delta_{c,d} = \bar{g}_{c,d} - \bar{g}_{\mathrm{NORM},d}$, where $\bar{g}_{\mathrm{NORM},d}$ is the mean gap of healthy subjects in the same decade. Since regression bias depends only on chronological age, it is shared by healthy and diseased subjects within each decade and is canceled by subtraction, leaving an estimate of the disease-specific effect. Confidence intervals for $\Delta_{c,d}$ are estimated using a nonparametric bootstrap by independently resampling the disease and NORM gap sets with replacement for 2,000 replicates and reporting the 2.5–97.5 percentile interval. Finally, the *pooled delta* summarizes each disease class as the count-weighted mean of $\Delta_{c,d}$ over the supported age range of 40–79 years, with confidence intervals obtained using the same bootstrap procedure across decades. Table 5 reports the metrics and three patterns stand out.

First, the raw gap reflects age bias rather than disease effects. It decreases with age for every class and becomes negative in the oldest decades. For example, MI declines from $+14.5$ years in the forties to -4.8 years in the seventies, which would incorrectly suggest that disease makes older hearts appear younger. The healthy baseline explains this pattern, decreasing from $+4.6$ to -8.8 years over the same range (bottom row of Table 5). Thus, raw gaps primarily capture age-related bias rather than pathology.

Second, the age-matched delta removes this baseline trend. Across the fifties, sixties, and seventies, all four classes converge to approximately $\Delta \approx 4$ years, with MI values of 4.5, 4.1, and 4.0, despite nearly twenty years of variation in the raw gap. This stability indicates that baseline subtraction eliminates age bias and isolates a disease-associated component. The forties remain an exception, with larger deltas (e.g., MI $+9.9$, CD $+5.8$) driven by small sample sizes ($n \leq 33$) and intervals that often include zero.

Third, pooled deltas are tightly clustered across diseases: STTC $+4.1$, MI $+4.6$, HYP $+3.3$, and CD $+4.6$. Their confidence intervals 95% overlap greatly, indicating that the gap distinguishes disease from health but not among disease classes. Therefore, the healthy-reference gap provides a sensitive but non-specific measure of pathology-associated aging across ages 40–79.

5. Conclusion

This work introduced LEAN, a lightweight and scalable hybrid network that pairs a shallow convolutional backbone with self-attention for ECG age estimation. The ECGNext backbone keeps the parameter count low, so the smallest variant, LEAN-Lite, exceeds the most compact network in the literature while using 42 times fewer parameters. A self-attention block recovers the long-range dependencies that the shallow backbone cannot capture on its own, and a multi-shot inference scheme averages predictions over several crops of a recording to refine the final estimate. LEAN comes as a family of variants, so practitioners can pick the configuration that best suits their hardware, trading inference time, VRAM usage, disk footprint, and multiply-accumulate cost against accuracy. The architecture also holds up on a demographically different population, retaining its accuracy when trained and evaluated on the Chapman dataset. Beyond age estimation, the healthy-reference analysis demonstrates that LEAN-derived age gaps capture a consistent pathology-associated aging signal, with diseased ECGs appearing approximately four years older than age-matched healthy recordings across multiple diagnostic classes. One limitation remains: the lighter variants lose accuracy on the tail of the age distribution, where rare and underrepresented ECG patterns are harder to fit. Even so, the overall design brings accurate ECG age estimation within reach of resource-constrained settings such as point-of-care diagnostics, where it can support the early detection and proactive management of cardiovascular disease.

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